

Visual Sim-to-Real Learning for Robotic Insertion under Geometric Variations: Application to Rebar Installation

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Abstract—Rebar insertion is among the most repetitive and physically demanding tasks on construction sites, and a contact-rich problem at 1.4 mm clearance. The parts, however, vary at two levels: a nominal design per structural member, and fabrication tolerance around each nominal design. Real-world data therefore has to be re-collected as designs and batches change. We present RebarSim, a visual sim-to-real system trained entirely in simulation. A privileged state-based teacher is trained with reinforcement learning over procedurally generated rebar geometries, then distilled into a multi-view student that maps raw RGB and proprioception directly to actions under extensive domain randomization. The student transfers to the real world zero-shot, seating rebars taken from a real factory production run in 91.3% of real-robot rollouts. Underlying that result, geometry diversity and pretraining both bring benefits. Training across a diverse set of nominal designs rather than one lifts the zero-shot success of both the teacher and the student on unseen designs, and the student policy outperforms a single-design specialist on that specialist’s own design. A pretrained student then adapts to a new design with 4–6 $\times$ fewer distillation samples than one trained from scratch. Visual sim-to-real transfer depends on appearance randomization and the Dagger mixture: removing either one sharply lowers success. Videos, code and task assets: <https://rebarsim.github.io>.

I. INTRODUCTION

On construction sites, rebar-cage assembly remains largely manual (Fig. 1). Workers repeatedly lift rebars, align them with narrow slots in support racks or other rebars, and insert them into place, often thousands of times per day. This repetitive work is costly, physically demanding, and associated with musculoskeletal injuries [1], while the construction industry faces a growing shortage of skilled labor [2]. Automating rebar insertion is therefore an important step toward scalable robotic rebar-cage assembly.



Fig. 1. Manual rebar insertion on site (left, middle): the worker lifts each rebar, then forces it down into the slots, thousands of times a day. Our robot setup for the same task (right).

Rebar insertion is also a challenging manipulation problem. Unlike conventional industrial insertion of a fixed, standardized part, a construction site encounters a family of

rebar geometries. Different structural members require different nominal designs, while fabrication imprecision further introduces variations in corner angles, cut lengths, out-of-plane twists, and hook geometry (Fig. 4). The rebar’s in-hand pose also varies from trial to trial. Across all of this variation, the robot must still achieve millimeter-scale alignment and resolve contact during insertion.

Existing approaches do not directly address this setting. Previous rebar insertion work [3] learns from real demonstrations, making it costly to scale training across geometric variation. Simulation makes such variation inexpensive through procedural generation and massively parallel experience collection [4], [5]. However, prior sim-to-real insertion methods that handle multiple part geometries [6], [7] rely on state-based policies and accurate object pose estimates at deployment. These assumptions are difficult to satisfy for fabricated rebars, whose geometry and in-hand pose are both uncertain. OmniReset [8] instead transfers an end-to-end RGB insertion policy from simulation to the real world without pose estimation. However, it is validated only on laboratory peg-in-hole tasks, and its generalization across object geometries is not evaluated.

This work therefore answers whether an end-to-end visual policy that generalizes across both sources of variation can be trained entirely in simulation and transferred to the real world. We introduce RebarSim, which extends the existing visual sim-to-real paradigm to this setting and learns insertion without real-world demonstrations or an intermediate object-pose estimation stage. To our knowledge, this is the first demonstration of an end-to-end visual sim-to-real policy for object insertion across variations in both object geometry and in-hand pose on a practical industrial task. RebarSim achieves 91.3% success over 150 real-world insertion trials spanning geometric variations and randomized in-hand poses. Simulation experiments further show zero-shot generalization to designs beyond the training distribution and fast adaptation to new designs through fine-tuning.

RebarSim follows a teacher-student paradigm [9], [10], [11]. We train a privileged state-based teacher with reinforcement learning over procedurally generated geometries and randomized object initializations, and distill it into a multi-view RGB policy using Dagger [12].

Our contributions are:

- 1) To our knowledge, we propose the first pipeline that learns an end-to-end visual sim-to-real policy that can perform object insertion across variations in both object geometry and in-hand pose, without real-world

demonstrations, validated on a practical rebar insertion task.

- 2) A study of how training-time geometry diversity affects generalization, showing that it improves robustness within the training range and reduces the adaptation required for geometries beyond it.
- 3) An empirical study of the visual sim-to-real recipe, showing that visual appearance randomization and the DAgger mixture are key ingredients for successful real-world transfer.

II. RELATED WORK

Sim-to-Real for Robotics Assembly. Robotic assembly is a long-standing problem in robot learning [13]. Many works learn from real-world teleoperated demonstrations, either by training a behavior cloning policy from scratch [14] or by fine-tuning a generalist vision-language-action (VLA) model [15]. These methods require a large number of real demonstrations and often generalize poorly beyond the demonstrated distribution. Simulation-based RL removes the need for real-world data, and domain randomization [16], [17], [18] and teacher-student distillation [9], [19], [11], [20], [21], [22] transfer the resulting policy to the real robot.

Previous works in sim-to-real robot assembly learning differ in what their policies observe. IndustReal [6] and AutoMate [7] policies consume 6D object poses as input, but require accurate real-world pose estimation with known CAD models. However, in our problem, fabricated rebars are not standard. In fact, almost no rebar matches its nominal drawing exactly, which weakens the assumption of prior works. Our policy instead reads raw RGB images. OmniReset [8] recently transferred RGB visual policies zero-shot on insertion tasks using a broad set of reset states and heavy domain randomization. However, it is limited to simple peg-in-hole tasks, while we address a practical construction task.

Automating Rebar Cage Assembly. Prior work on automating rebar cage assembly has explored primitives such as rebar grasping, placing, and tying. Most of the work addresses *tying*. Typically, the pipeline first recognizes the intersection, then estimates its explicit pose, and plans the tying gun’s trajectory, often on gantry- or track-mounted hardware [23], [24], [25]. These works assume that rebars are already in place on the rack and that their poses can be recovered by explicit geometric modeling. Our work focuses on the placing primitive.

The work closest to ours [3] performs insertion of a tight-fit rebar slot on a mobile manipulator by combining visual servoing with imitation learning over prompt-segmentation masks [26]. However, it focuses on similar rebar shapes, and the robustness of the policy is not tested against large rebar shape variation. Moreover, it is costly to scale the method to various rebar geometries. RebarSim addresses this gap. We learn the same task from simulation alone, so geometric variation is covered by procedurally generated training rebars.

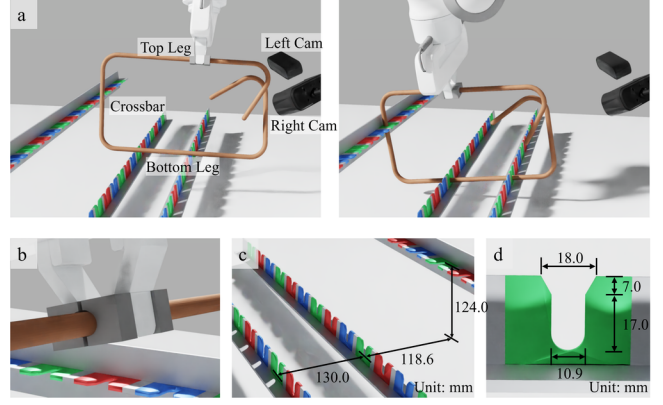


Fig. 2. Task overview. (a) the robot inserts the rebar it holds into a three-slot rack, guided by the left and right cameras and its proprioception. (b) the gripper fingers. (c) rack dimensions. (d) slot dimensions.

III. METHOD

Our RL environment and training are implemented in Isaac Lab [4]. The robot begins each episode holding a rebar (a steel rod bent into a closed rectangular loop) and must seat it in three designated slots of a support rack with 1.4 mm clearance (Fig. 2). The gripper holds one leg of the loop, which we call the top leg; the opposite leg (the bottom leg) seats in two low slots and the crossbar, the branch opposite the hooks, in a third, higher slot (Fig. 2). Here, we assume that the rebar is already in hand, with a random gripping pose within a reasonable range. Rack slots are standardized industrial hardware, so the geometry is fixed and known. Let δ_k denote the rebar’s remaining distance to its seated pose at slot k , measured along the insertion direction. An episode succeeds when $\max_k \delta_k \leq \delta^* = 5$ mm.

RebarSim has three development stages (Fig. 3): a privileged *teacher* learning, a visual *student* distillation, and zero-shot *real-world deployment*. The policy emits a task-space delta pose $\mathbf{a}_t \in \mathbb{R}^6$ of the fingertip frame at 10 Hz. Joint torques are computed with a task impedance controller (Sec. III-C). The teacher $\pi_{\text{teacher}}(\mathbf{a}_t \mid o_t^{\text{priv}})$ acts on privileged simulator state, and the student $\pi_{\text{student}}(\mathbf{a}_t \mid o_t^{\text{vis}})$ reads $o_t^{\text{vis}} = (\mathbf{I}_t^{1:K}, o_t^{\text{prop}})$: $K=8$ calibrated RGB views and the proprioception o_t^{prop} defined in Sec. III-D.

A. Simulation with Geometric Variation

We cast the task as a finite-horizon POMDP whose dynamics are parameterized by a task instance

$$\xi = (g, \mathbf{s}_0) \sim p(\xi), \quad g = (d, \epsilon) \in \mathcal{G}, \quad (1)$$

resampled at every reset in simulation. Here g is the rebar geometry, a nominal design d with fabrication deviation ϵ , and $\mathbf{s}_0$ is the initial scene state. This subsection describes how we generate $\mathcal{G}$, and Sec. III-B how we generate $\mathbf{s}_0$.

We scope $\mathcal{G}$ to *stirrups* (the transverse rebars wrapping the longitudinal steel of essentially every concrete beam, column and pile cage, and among the most numerous rebar shapes on site) in the closed rectangular form of Fig. 4.

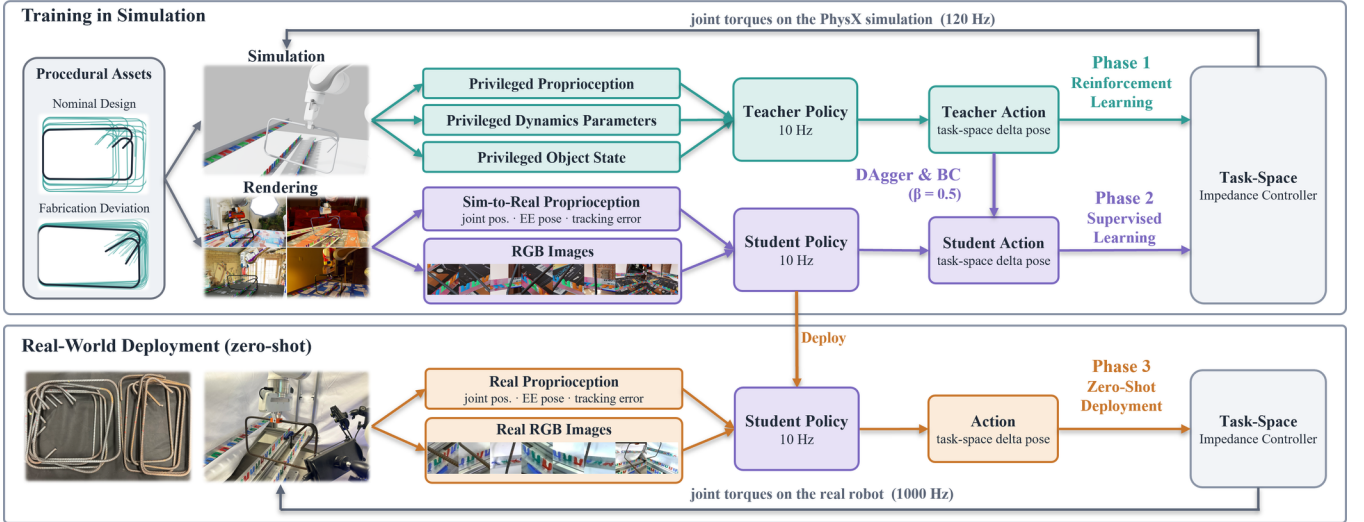


Fig. 3. Overview of RebarSim. *Training in simulation*: a procedural generator generates the rebar assets. A privileged teacher is trained with RL (Phase 1, teal) and supervises a student that sees only rendered RGB and proprioception (Phase 2, purple). *Deployment*: the same student, frozen, on the real cameras (Phase 3, orange).

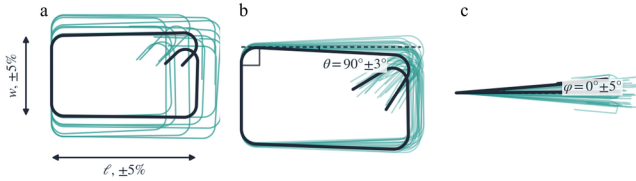


Fig. 4. The rebar generator’s parameters. (a) the lengths ℓ and w . (b) the corner angle θ . (c) the out-of-plane twist φ .

Rebar Generator. We design our rebar generator to cover a wide range of rebar geometries in training. A nominal design is $d = (\ell, w)$, the leg length and the width of the crossbar. Fabrication then perturbs it by $\epsilon = (\epsilon_\ell, \epsilon_w, \{\theta_i\}_{i=1}^3, \{\varphi_i\}_{i=1}^3, h, \alpha)$: the leg length ℓ and the width w each at $\pm 5\%$, each of the three straight branches (two legs and the crossbar) at corner angle $\theta_i = 90^\circ \pm 3^\circ$ and out-of-plane twist $\varphi_i = 0^\circ \pm 5^\circ$, and the two hooks sharing a length $h \in [20, 100]$ mm and an angle $\alpha \in \{45^\circ, 60^\circ, 90^\circ\}$. The parameters and their ranges come from standard fabrication practice and from the rebars we measured. Our experiments use #3 rebar, a common size on site, with the diameter held at its 9.5 mm nominal, since rolling tolerance is far tighter than cutting and bending tolerance.

Physics Modeling. Both rebars and the rack use signed-distance-field (SDF) collision detection. This work studies the insertion skill itself, with the target fixed to the center slot group, so only that group carries SDF collision geometry. Selecting the target group, for example from a color or position cue in the observation, is left to future work. We simulate the in-hand grasp as a revolute hinge along the hand’s x-axis with friction. This simplification reduces simulation instability from thin rebars but still allows sim-to-real transfer.

B. Reset-State Bank Generation

Reset states covering every stage of a task let on-policy RL master long-horizon, contact-rich skills with simple rewards [8]. Ours randomize the outcome of a preceding grasp: where the rebar sits in the hand, and where the hand sits relative to the rack. We precompute two families of reset states, mixed 1:1 during training, at 5,000 states per rebar variant.

Free-space approach states place the grasped rebar uniformly in a box anchored at the variant’s success pose and expressed in the rack frame (up to 5 cm horizontally withdrawn from the seated pose, ± 3 cm lateral, up to 5 cm above), with per-axis orientation jitter of ± 0.17 rad.

Partially inserted states are generated by *physics back-out*: with gravity off, randomized wrenches push the rebar from the success pose along its low-resistance directions, and every still-engaged intermediate pose is recorded. We reject-sample to $\approx 1/3$ each for 3, 2 and 1 engaged slots, and drop states that are already seated.

Each recorded state additionally carries a randomized arm posture, grasp and rack placement. IK over this confined workspace returns configurations from a narrow band, so each solution is perturbed inside the nullspace of the task Jacobian, moving the elbow but not the rebar. The in-hand roll (± 0.12 rad), the grip slide along the top leg’s axis (± 3 cm) and the rack pose (± 3 cm, ± 0.15 rad) are sampled once at generation time, at ranges that cover the range an upstream picking stage such as [3] produces.

C. Privileged Teacher Training

Teacher formulation. We formulate the teacher as a state-based RL policy $\pi_{\text{teacher}}(\mathbf{a}_t \mid \mathbf{o}_t^{\text{priv}})$ trained with PPO [27]. Actor and critic are separate MLPs of width [512, 256, 128, 64]. The privileged observation stacks the previous action, proprioception, the rebar’s roll in the grip, and the rebar and

TABLE I
REWARD TERMS AND WEIGHTS.

Term	Expression	λ_i
Alignment	$\frac{1}{2} \left[e^{-\Delta p_t / \sigma_p} + e^{-\Delta \psi_t / \sigma_a} \right]$	0.1
Success	$\mathbf{1}[\max_k \delta_k \leq \delta^*]$	1.0
Contact force	$\frac{1}{T F_{\text{ref}}} \sum_j \text{clamp}(\ F_{t,j}\ - F_0, 0, 200 \text{ N})$	-0.05
Action mag.	$\ \mathbf{a}_t\ ^2$	-10^{-4}
Action rate	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ ^2$	-10^{-3}
Arm joint vel.	$\ \dot{\mathbf{q}}_t\ ^2$	-10^{-2}
Abnormal arm	$\mathbf{1}[\dot{q} > 2 \dot{q}_{\text{max}}]$	-100

rack poses in the rack frame over the last five control steps, plus two task-specific terms. $\mathbf{x}_t^{\text{seg}}$, the pose of the bottom leg and of the crossbar (Fig. 2) in the rack frame, exposes the sampled geometry as segment poses (midpoint and direction) rather than as shape parameters. $\mathbf{f}_t$, the peak rebar-rack contact force and its excess over the penalty threshold, lets the policy modulate contact rather than only being penalized for it.

Action Space. The action $\mathbf{a}_t = (\Delta \mathbf{x}_t, \Delta \omega_t) \in \mathbb{R}^6$ is a Cartesian pose delta relative to the *current* fingertip pose, scaled by 0.04 m and 0.06 rad per step and tracked over the seven arm joints by a task impedance controller,

$$\boldsymbol{\tau} = \text{clamp}(J^\top [K_p \mathbf{e}_t - K_d \mathbf{v}]), \quad (2)$$

where $\mathbf{e}_t$ is the fingertip pose tracking error. Note that the clamp acts on the joint torque, not on $\mathbf{a}_t$. The critic additionally observes privileged quantities: remaining episode time, velocities, and the dynamics parameters sampled by the randomization below.

Reward Design. The reward is a fixed weighted sum $r_t = \sum_i \lambda_i r_i$ of the terms in Table I, shared unchanged by every geometric variant, with no staged shaping. The only schedule applies to the contact term, whose magnitude is ramped up linearly over the first 300 iterations.

Here Δp_t and $\Delta \psi_t$ are the position and orientation deviation of the rebar from its per-variant seated pose, computed analytically in the rack frame, with scales $\sigma_p=0.10$ m and $\sigma_a=0.60$ rad. The contact term penalizes only the rebar-rack force $F_{t,j}$ above $F_0=30$ N, averaged over the $T=12$ physics substeps j per control step and normalized by $F_{\text{ref}}=100$ N, so that pressing hard is discouraged.

Domain Randomization. Dynamics randomization is anchored at system-identified nominal values [17], [11]: arm friction, armature, and task-impedance gains are jittered by $\pm 20\%$ around the SysID point, which is provided with the released controller. Surface friction is drawn per environment: static and dynamic μ from $U(1.0, 2.0)$ and $U(0.9, 1.9)$ for both the rebar and the rack, and from $U(0.3, 1.2)$ and $U(0.2, 1.0)$ for the gripper pads.

D. Visual Student Distillation

The student π_{student} observes proprioception and eight RGB views: six close-up views centered on the slots, plus two

wide views that keep the whole rebar in frame. Sec. III-E describes how these views are obtained in simulation and on the real robot.

Dagger-BC Distillation. We distill the privileged teacher into the student with a hybrid of online DAgger [12] and behavior cloning (BC) [28]. Both share the same regression objective, computed over a mixture of the observation distributions induced by teacher and student rollouts:

$$\mathcal{L}_{\text{distill}} = \mathbb{E}_{o_t \sim \rho^o} \left[\left\| \pi_{\text{teacher}}(o_t^{\text{priv}}) - \pi_{\text{student}}(o_t^{\text{vis}}) \right\|_2^2 \right], \quad (3)$$

where $\rho^o = \beta \rho_{\pi_{\text{teacher}}}^o + (1 - \beta) \rho_{\pi_{\text{student}}}^o$ mixes the two rollout distributions with ratio β , and o_t^{priv} , o_t^{vis} are the privileged and visual observations of the same simulator state. We use $\beta = 0.5$, following the ratio ablation of [28].

Network Architecture. One ImageNet-pretrained ResNet-18 encodes all eight views, and a learnable per-view embedding marks which view a feature came from. The features are concatenated with the student’s proprioception $o_t^{\text{prop}} = [\mathbf{q}_t, \mathbf{x}_t^{\text{ce}}, \mathbf{e}_t]$ (the nine joint positions of the arm and gripper fingers, the fingertip pose in the robot base frame, and the fingertip tracking error, 21 dimensions carrying no previous action and nothing about the rebar or the rack) over a four-step history, and mapped to the action by an MLP with two hidden layers of 512 ReLU units and a linear output layer.

Domain Randomization. Because the student reads pixels directly, distillation randomizes everything that changes how the scene looks. Dome lighting is resampled from a pool of HDRI environment maps. The rebar, the rack, the table and the robot each receive an independently sampled texture and surface material (color, roughness and metallic). Camera poses are jittered around the calibrated real setup: ± 1 cm on the optical center of all eight views, and a further $\pm 3^\circ$ of orientation on the two wide ones. Intrinsic are held at their calibrated values. Geometry and dynamics randomization continue from teacher training. Fig. 5 (top) shows the spread of scenes this produces.

E. Real-World Deployment

Control. We use a Franka Research 3 robot. Policy actions are executed by a task impedance controller of Sec. III-C, whose torque is interpolated to 1 kHz in the real world. In the real world, the rebars used in our experiments weigh about 0.7 kg; we compensate this gravity load on the end effector manually, and leave automatic compensation to future work.

Perception. Fig. 3 (bottom) shows our robot setup: two ZED Mini stereo cameras are mounted to see the rack and the robot’s workspace. We recover their poses from a calibration marker and place the simulated cameras correspondingly.

In our simulation the eight views are rendered by eight cameras. In the real world, each of the two rectified left images is cut into fixed bounding-box crops, three close-up views centered on the slots plus one wide view (Fig. 5, bottom), and each crop is warped with a homography into a virtual camera that faces its target as the corresponding simulated camera does.

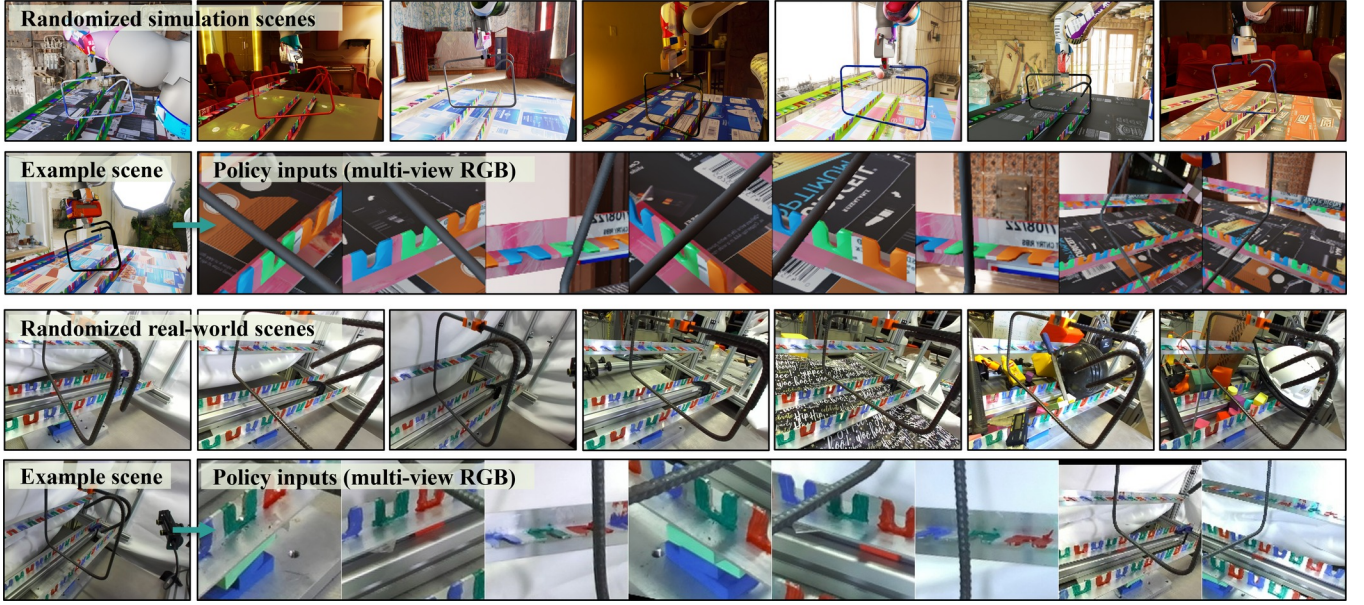


Fig. 5. **Training scenes in simulation (top) and evaluation scenes on the real robot (bottom).** The student never sees a real image while it is trained: the top block is the distribution it is distilled on, the bottom block is the real-world setup it is deployed in. Each block pairs the spread of randomized scenes with one example scene and the multi-view RGB the policy actually reads in it.

IV. EXPERIMENTS

We design our experiments to answer the following questions:

- Q1** Does RebarSim generalize across rebar geometry, in simulation and on the real robot, with no real-world data?
- Q2** How does geometry randomization during training affect final performance, and how does it affect the speed of adapting to novel geometries?
- Q3** Which design choices of the visual sim-to-real recipe are critical in practice?

Simulation Training. All policies are trained on four NVIDIA L40 GPUs. The teacher trains state-only in 16,384 environments per GPU. The student renders its eight 150×150 camera views with tiled rendering at 74 environments per GPU. We evaluate on three sets of nominal designs. $\mathcal{D}_{\text{train}}$ is the 3×3 grid $\ell \in \{290, 350, 400\} \text{ mm} \times w \in \{195, 240, 290\} \text{ mm}$. $\mathcal{D}_{\text{interp}}$ is four unseen designs inside that grid, $\ell \in \{320, 375\} \text{ mm} \times w \in \{217.5, 265\} \text{ mm}$. $\mathcal{D}_{\text{extrap}}$ is the four corners of $\ell \in \{260, 430\} \text{ mm} \times w \in \{150, 320\} \text{ mm}$, which lie outside the grid on both axes. Each design carries 20 variants, giving 180, 80 and 80, with ϵ drawn throughout from the tolerance ranges of Sec. III-A, so the design d is the only factor that differs between the sets. Every simulation success rate we report is measured with the deterministic policy over 4096 episodes for the teacher and 2304 for the student, with rebar variants drawn uniformly at random from the evaluated set. Simulated and real evaluations both start from a held-out evaluation bank of free-space states (Sec. III-B).

Real-world Evaluation. Real rollouts cover two designs, $\mathcal{D}_1$ ($\ell = 350, w = 195 \text{ mm}$) and $\mathcal{D}_2$ ($\ell = 290, w = 290 \text{ mm}$).

Both are grid points of $\mathcal{D}_{\text{train}}$, and for each we use a batch of rebars from a real factory production run. The rebars deviate markedly from their nominal geometry: corner angles reach their 3° tolerance limit and out-of-plane twists their 5° limit (Sec. III-A). Both are scored in simulation *and* on the real robot, so their gap is a sim-to-real drop of one policy on one design.

Task Metric. We report the success rate, under the $\max_k \delta_k \leq \delta^*$ criterion of Sec. III applied identically in our simulated and real-robot evaluations.

A. Generalization across Geometric Variations (Q1)

Table II reports RebarSim in simulation and on the real robot. In simulation, it scores 96.1% on the nine designs it trained on and 96.6% on $\mathcal{D}_{\text{interp}}$, the four unseen designs held out inside that grid. That performance transfers to the real robot, where it seats the rebar in 137 of 150 rollouts (91.3%): 68/75 on $\mathcal{D}_1$ and 69/75 on $\mathcal{D}_2$, against 95.4% and 96.8% for the same policy in simulation. We observed no pronounced difference in success rate between individual rebars. The dominant failure mode is discussed in Sec. IV-E, with a more detailed account on the project website.

Background robustness. Fig. 6 examines how the visual background affects the policy on the real robot. Across 30 rollouts in five deployed scenes (a construction lab, two re-textured and two cluttered), it seats 25 rebars, against 6 of 6 in the clean lab. Success degrades rather than collapses: the two hardest scenes, a marble backdrop and a heavily cluttered scene with toys, tools and cables, still seat 4 of 6.

B. Effect of Geometry Randomization (Q2)

Rebar geometry factors as $g = (d, \epsilon)$ (Sec. III), the nominal design and the fabrication deviation. We build four

TABLE II

REBARSIM’S SIMULATED AND REAL ROLLOUT RESULTS ON DIFFERENT REBAR GROUPS. ON THE REAL ROBOT, $\mathcal{D}_1$ AND $\mathcal{D}_2$ EACH CONTRIBUTE FIVE REBARS DRAWN AT RANDOM FROM THEIR GROUP, FIFTEEN ROLLOUTS PER REBAR, WITH THE REBAR’S INITIAL POSE, THE GRASP POSE AND THE RACK POSE RANDOMIZED AS IN SIMULATION. REAL ENTRIES ARE THE SUCCESS RATE (%), WITH SUCCESSES/ROLLOUTS IN PARENTHESES.

Evaluation set	Sim SR	Real
$\mathcal{D}_{\text{train}}$ (9 designs)	96.1	– (sim only)
$\mathcal{D}_{\text{interp}}$ (4 unseen designs)	96.6	– (sim only)
$\mathcal{D}_{\text{extrap}}$ (4 designs outside grid)	77.6	– (sim only)
$\mathcal{D}_1$	95.4	90.7 (68/75)
$\mathcal{D}_2$	96.8	92.0 (69/75)
all real rollouts	–	91.3 (137/150)

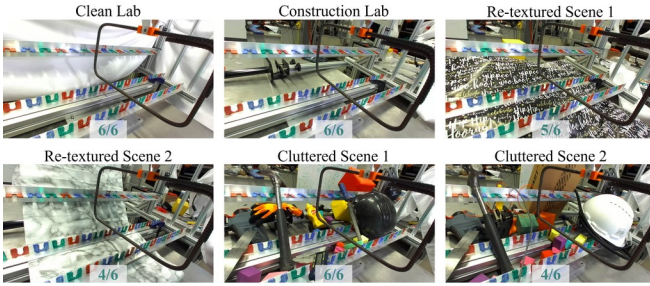


Fig. 6. Real-robot success across backgrounds. The six rollouts are the same six everywhere (two rebars drawn at random from $\mathcal{D}_1$, three initial EE poses and rebar-in-hand poses each, held fixed as the scene changes), so the panels differ in background alone. The five deployed scenes pool to 25/30 against 6/6 in the clean lab, top left.

training cases as a 2×2 factorial over which component of g is randomized. All four train on a set that contains $\mathcal{D}_1$ ’s nominal design and all are scored on $\mathcal{D}_1$: neither (that design as a single exact rebar), ϵ only (that design across its tolerances, i.e. $\mathcal{D}_1$ itself), d only (the nine exact designs of the training grid, which include it), and both (those nine across their tolerances, the set RebarSim trains on). We train a teacher and its distilled student for each, and report the student (Fig. 7).

Fig. 7 separates the two main effects. With ϵ held fixed, randomizing the design d lifts success from 79.4% to 95.9%. With d held fixed, randomizing the fabrication deviation ϵ lifts it to 89.8%. Both matter, and randomizing both is best, at 96.6%. Design diversity has the larger main effect (16.5 points against 10.4) and the larger marginal effect: adding d on top of ϵ gains a further 6.8 points, adding ϵ on top of d only 0.7. Diversity along d therefore transfers to ϵ while the reverse does not: a policy trained across the nine exact designs already handles fabrication deviations it was never trained on, whereas one trained on a single design across its full tolerance range still trails the nine-design policy by 6.8 points on the very design it trained on. Design-level diversity also does not reduce performance on any single design: the

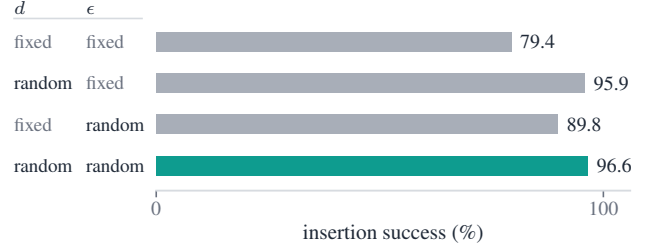


Fig. 7. Student evaluation on $\mathcal{D}_1$ under the four training cases. For each training case, we report the success rate of its distilled student’s best checkpoint on $\mathcal{D}_1$.

nine-design policy is the best of the four on this one design, a conclusion similar to the object-generalization result reported for grasping in [28].

C. Adaptation Speed to Novel Geometries (Q2)

We fine-tune on $\mathcal{D}_{\text{extrap}}$ at both levels of the pipeline. **Teacher level:** three initializations — RebarSim (teacher randomizing both d and ϵ), the d -fixed and ϵ -random teacher of Sec. IV-B, and random weights, i.e. from scratch — are fine-tuned on the new designs and their tolerance ranges under one common budget of 1.0B samples, three seeds each, and compared by the number of samples needed to reach 90% success. **Student level:** the adapted RebarSim teacher is then distilled into students initialized from the pretrained RebarSim student, from the pretrained d -fixed student, and from scratch, three seeds each. All three distill from that one teacher, so the student’s starting weights are the only difference.

At the teacher level, shown in Fig. 8 (a), training across diverse nominal designs raises zero-shot success on the four unseen designs from 55.4% for single-design pretraining to 92.6%, against 0.0% from scratch, so the three initializations rank by how many designs they saw in pretraining. Diversity also shortens adaptation: RebarSim starts above the 90% mark (fine-tuning adds a further 5.6 points), whereas the d -fixed teacher needs 77M samples to reach it and training from scratch 667M. All three teachers converge to 96–98% within the shared budget. At the teacher level, diversity therefore determines how many samples a new design takes, not the final performance on it.

At the student level, shown in Fig. 8 (b), the dominant factor is whether the student was pretrained at all. Zero-shot, the RebarSim student already reaches 77.6% on $\mathcal{D}_{\text{extrap}}$, against 55.1% for the d -fixed student. When adaptation is needed, the RebarSim and d -fixed students reach 85% after 0.19M and 0.26M distillation samples, 6.2 $\times$ and 4.5 $\times$ fewer than the 1.17M a student trained from scratch needs, and they plateau at 90–93% over their last four evaluations against the from-scratch student’s 81–85%. Pretraining therefore both reduces the distillation samples a new design needs and raises the final success rate.

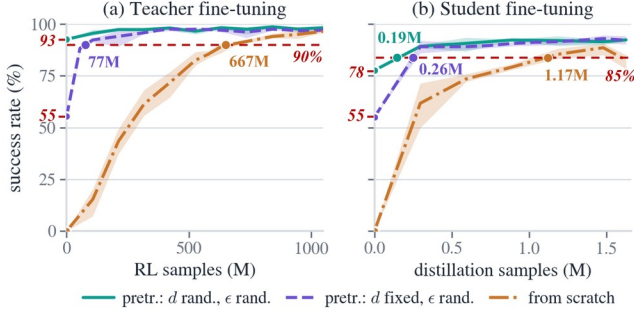


Fig. 8. Adaptation to $\mathcal{D}_{\text{extrap}}$, the four designs outside the training grid. Labels indicate which components of the rebar geometry $g = (d, \epsilon)$ the pretraining randomized. The x -axis counts only adaptation samples, so each curve starts at its zero-shot success, marked on the left axis. The line is the mean over three seeds, the band their min-max. (a) teacher RL fine-tuning, dots at samples-to-90%. (b) student distillation, dots at samples-to-85%.

TABLE III

TRAINING-RECIPE ABLATIONS, EVALUATED IN SIMULATION ON THE NINE DESIGNS OF $\mathcal{D}_{\text{TRAIN}}$. ALL ROWS ABLATE DISTILLATION WITH THE TEACHER, RESET BANKS AND ASSETS HELD FIXED.

Recipe	Sim SR (%) $\uparrow$
RebarSim (full recipe)	96.1
w/o appearance randomization	2.8
w/o viewpoint randomization	90.2
pure BC (no DAgger)	68.9

D. Visual Sim-to-Real Recipe Ablations (Q3)

We ablate three ingredients of the recipe, removing or replacing each one in turn: appearance randomization, viewpoint randomization and the DAgger mixture. Table III reports the results in simulation on $\mathcal{D}_{\text{train}}$.

Appearance randomization is indispensable. The student trained without it still solves its own fixed-appearance environment (91.1), but collapses to 2.8 under the randomized appearances the deployed policy is scored on. The other two matter less. Viewpoint randomization is worth 5.9 points. Replacing the DAgger mixture with pure BC costs 27.2 points, a drop we attribute to the covariate shift of a student that is never asked to act from its own states. The mixture keeps the teacher-to-student drop small on this task.

E. Observed Behaviors and Failure Modes

Emergent recovery. For rebar insertion, the most common failure is the rebar colliding with the rack near the slots and getting stuck there. Such contact states are many and varied, so [3] collects dedicated failure-recovery demonstrations to handle them, which takes substantial time and effort. In our real rollouts, about 20% of the insertions did not seat the rebar on the first attempt: the rebar first collided with the rack around the slots, and the policy backed it out and re-attempted the insertion until the rebar seated (video on the project site). This back-out-and-retry behavior is not programmed anywhere in the policy or the controller. It

emerges from RL training with a reward that has no retry term (Table I).

Failed recovery. In some rollouts the policy does not back the rebar out after the collision but keeps rotating it to slide it into the slots. The grasp cannot transmit enough torque to overcome the contact friction, so the continued rotation produces a growing in-hand roll shift until the rebar can no longer be recovered. We attribute this to contact differences between simulation and the real world, which let the policy learn a behavior that is harder to execute on the real robot.

V. DISCUSSION AND CONCLUSIONS

We built and deployed RebarSim, a visual sim-to-real system for contact-rich insertion under geometric variation, instantiated on rebar insertion. Trained entirely in simulation over procedurally generated geometric variants and deployed directly from raw camera images, it seats rebars taken from a real factory production run at a 91.3% success rate over 150 real-robot rollouts, with no real-world data anywhere in the pipeline. Our ablations trace that result to a few design choices. Geometry diversity and pretraining both bring benefits: training across nine nominal designs rather than one raises success on the training designs and lifts zero-shot success outside them from 55.4% to 92.6%, while pretraining of either kind makes a new design sample-efficient to acquire, at 4–6 $\times$ fewer distillation samples than starting from scratch. The visual recipe rests on appearance randomization, without which the student does not transfer at all, and the DAgger mixture, worth 27.2 points over pure behavior cloning.

The deployed system has two limitations, and both point to replacing a hand-designed component with one learned in simulation. First, RebarSim assumes the rebar is already grasped, and acquiring that grasp is left to a preceding module or policy, so we demonstrate only the insertion stage. We intend to train that stage in simulation with RL as well, and compose the two into a single grasp-and-insert pipeline. Second, the student’s viewpoints are fixed at design time: its eight views are fixed bounding-box crops (Sec. III-E) laid out for the target slot group, and other slot groups on the rack are not considered. Mounting the robot on a mobile base could keep the cameras aligned with whichever slot group is the target, but this limits how conveniently the system can be deployed. An active-vision mechanism, letting the policy choose which region of the image to resolve, would make that choice part of the learned policy.

REFERENCES

- [1] M. S. Forde, L. Punnett, and D. H. Wegman, “Prevalence of musculoskeletal disorders in union ironworkers,” *Journal of Occupational and Environmental Hygiene*, vol. 2, no. 4, pp. 203–212, 2005.
- [2] Associated General Contractors of America and National Center for Construction Education and Research, “2025 workforce survey analysis,” Tech. Rep., 2025. [Online]. Available: www.agc.org
- [3] T. Sun, B. Han, J. Wu, S. Rusinkiewicz, and Y. Shao, “Mobile robotic rebar cage assembly via imitation learning,” *Automation in Construction*, vol. 181, Art. no. 106671, 2026.
- [4] M. Mittal *et al.*, “Orbit: A unified simulation framework for interactive robot learning environments,” *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023.

- [5] Y. Narang *et al.*, “Factory: Fast contact for robotic assembly,” in *Robotics: Science and Systems (RSS)*, 2022.
- [6] B. Tang *et al.*, “IndustReal: Transferring contact-rich assembly tasks from simulation to reality,” in *Robotics: Science and Systems (RSS)*, 2023.
- [7] B. Tang *et al.*, “AutoMate: Specialist and generalist assembly policies over diverse geometries with declarative assembly specifications,” in *Robotics: Science and Systems (RSS)*, 2024.
- [8] P. Yin, T. Westenbroek, Z. Zhang, J. Tran, I. Dagnino, E. Shilamkar, N. Mbiziwo-Tiapo, S. Bagaria, X. Liu, G. Mullins, A. Kolobov, and A. Gupta, “Emergent dexterity via diverse resets and large-scale reinforcement learning,” *arXiv preprint arXiv:2603.15789*, 2026.
- [9] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning quadrupedal locomotion over challenging terrain,” *Science Robotics*, vol. 5, no. 47, 2020.
- [10] T. Chen, J. Xu, and P. Agrawal, “A system for general in-hand object re-orientation,” in *Conf. on Robot Learning (CoRL)*, 2021.
- [11] A. Handa *et al.*, “DeXtreme: Transfer of agile in-hand manipulation from simulation to reality,” in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2023.
- [12] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2011.
- [13] M. A. Lee, Y. Zhu, K. Srinivasan, P. Shah, S. Savarese, L. Fei-Fei, A. Garg, and J. Bohg, “Making sense of vision and touch: Self-supervised learning of multimodal representations for contact-rich tasks,” in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2019.
- [14] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Robotics: Science and Systems (RSS)*, 2023.
- [15] M. J. Kim *et al.*, “OpenVLA: An open-source vision-language-action model,” in *Conf. on Robot Learning (CoRL)*, 2024.
- [16] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” in *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2017.
- [17] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-real transfer of robotic control with dynamics randomization,” in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2018.
- [18] I. Akkaya *et al.*, “Solving Rubik’s cube with a robot hand,” *arXiv preprint arXiv:1910.07113*, 2019.
- [19] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Robotics: Science and Systems (RSS)*, 2021.
- [20] T. Chen, M. Tippur, S. Wu, V. Kumar, E. Adelson, and P. Agrawal, “Visual dexterity: In-hand reorientation of novel and complex object shapes,” *Science Robotics*, vol. 8, no. 84, 2023.
- [21] T. G. W. Lum *et al.*, “DextrAH-G: Pixels-to-action dexterous arm-hand grasping with geometric fabrics,” in *Conf. on Robot Learning (CoRL)*, 2024.
- [22] T. He *et al.*, “HOVER: Versatile neural whole-body controller for humanoid robots,” in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2025.
- [23] J. Jin, W. Zhang, F. Li, M. Li, Y. Shi, Z. Guo, and Q. Huang, “Robotic binding of rebar based on active perception and planning,” *Automation in Construction*, vol. 132, Art. no. 103939, 2021.
- [24] S. Fan, M. Liu, H. Li, H. Liang, Y. Yuan, and Y. Wang, “OpenTie: Open-vocabulary sequential rebar tying system,” in *IEEE Int. Conf. on Automation Science and Engineering (CASE)*, 2026.
- [25] M. Momeni, J. Relefors, A. Khatry, L. Pettersson, A. V. Papadopoulos, and T. Nolte, “Automated fabrication of reinforcement cages using a robotized production cell,” *Automation in Construction*, vol. 133, Art. no. 103990, 2022.
- [26] N. Ravi *et al.*, “SAM 2: Segment anything in images and videos,” *arXiv preprint arXiv:2408.00714*, 2024.
- [27] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [28] T. He *et al.*, “VIRAL: Visual sim-to-real at scale for humanoid loco-manipulation,” *arXiv preprint arXiv:2511.15200*, 2025.